

REAL ESTATE PLATFORM

USE CASE FLOW DOCUMENT

System Actors · Use Cases · Interaction Flows · Exception Handling

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TABLE OF CONTENTS

- 1. Document Overview & Actor Summary
- 2. Use Case Diagram — System Overview
- 3. Actor Definitions

SECTION A — CUSTOMER USE CASES

- UC-C01 Register & Create Account
- UC-C02 Login & Manage Profile
- UC-C03 Browse & Search Properties
- UC-C04 View Property Details
- UC-C05 Save Property to Wishlist
- UC-C06 Submit Property Inquiry / Lead
- UC-C07 Contact an Agent Directly
- UC-C08 Track Inquiry / Lead Status

SECTION B — AGENT USE CASES

- UC-A01 Login to Agent Portal
- UC-A02 View & Manage Assigned Leads
- UC-A03 Update Lead Status & Progress
- UC-A04 Log Call & Communication Activity
- UC-A05 Schedule Property Viewing
- UC-A06 Manage Customer Profile & Notes
- UC-A07 Manage Property Listings
- UC-A08 View Personal Performance Dashboard

SECTION C — ADMIN USE CASES

- UC-AD01 Login to Admin Portal
- UC-AD02 Manage Property Listings (Full CRUD)
- UC-AD03 Publish / Unpublish Listings
- UC-AD04 View & Manage All Leads
- UC-AD05 Assign / Reassign Leads to Agents
- UC-AD06 Create & Manage Users
- UC-AD07 Manage Roles & Permissions
- UC-AD08 Manage Website Content (CMS)
- UC-AD09 View Reports & Analytics
- UC-AD10 View Audit Logs

SECTION D — SUPER ADMIN USE CASES

- UC-SA01 System Configuration & Settings
- UC-SA02 Manage Integration Settings

SECTION E — SYSTEM / AUTOMATED USE CASES

- UC-SY01 Auto Lead Assignment
- UC-SY02 Send Automated Notifications
- UC-SY03 Escalate Uncontacted Lead

- 6. Cross-Cutting Concerns
- 7. Use Case Relationship Summary

1. DOCUMENT OVERVIEW

This document defines the complete Use Case Flow for the Real Estate Platform. It covers every interaction between actors (users and systems) and the platform across all three portals: the Public Website, the Agent Portal, and the Admin Portal.

Each use case includes: Actor, Description, Preconditions, Step-by-step Main Flow, Alternate/Exception Flows, and Postconditions. Use cases are grouped by actor for clarity.

Actor Summary

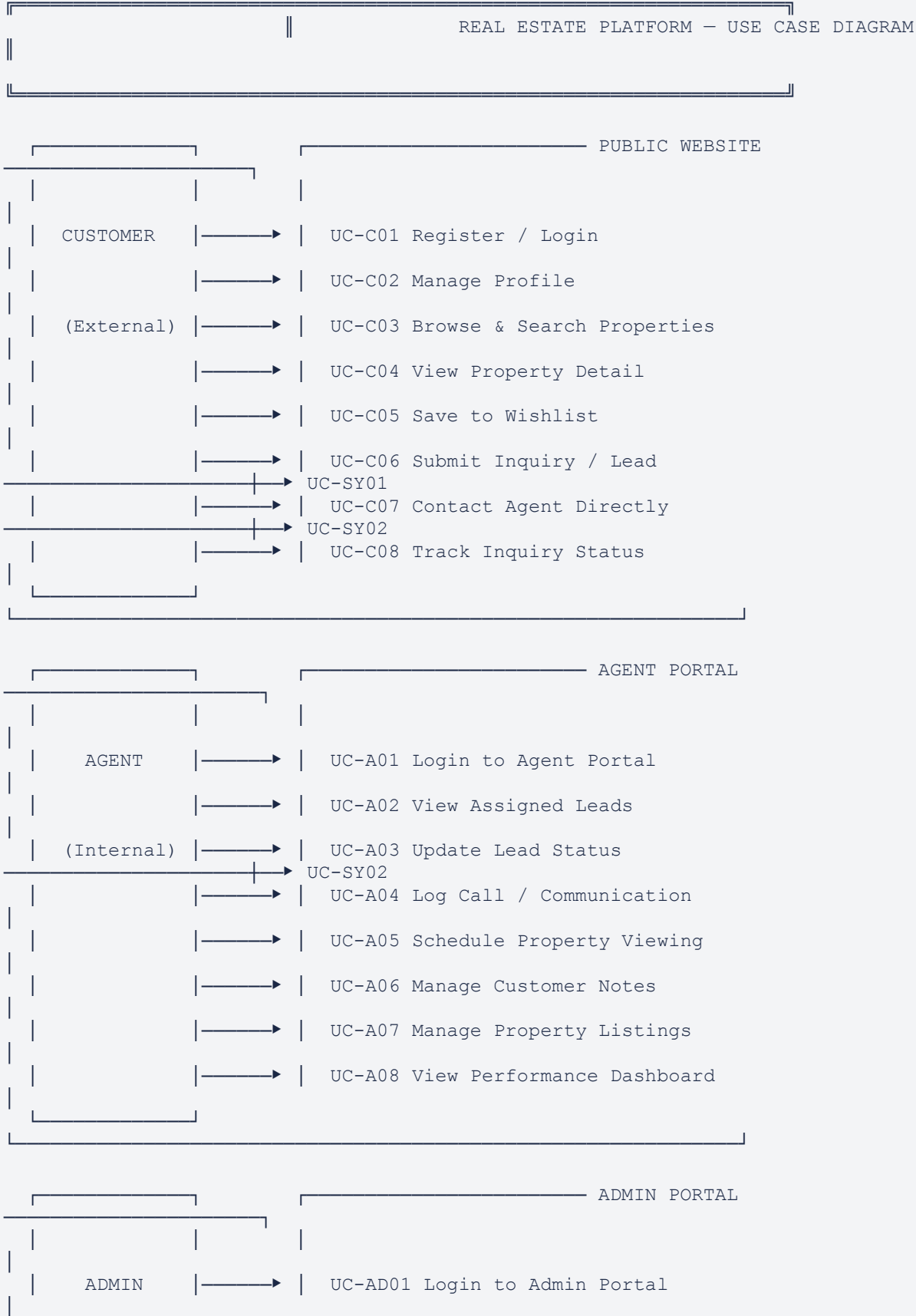
Actor	Type	Description
Customer	External User	End-users who visit the public website, browse listings, and submit inquiries/leads.
Agent	Internal User	Sales agents who receive assigned leads, manage customer interactions, and update deal progress.
Admin	Internal User	Operations administrators who manage properties, users, leads, content, and reports.
Super Admin	Internal User	Full system authority — manages role permissions, integrations, and all configuration settings.
Email System	System Actor	Automated email delivery system (SendGrid/Postmark) triggered by platform events.
SMS System	System Actor	Automated SMS/WhatsApp gateway (Twilio) triggered by platform events.
Ext. Portals	External System	Third-party property portals (Bayut, Property Finder) that receive syndicated listings.

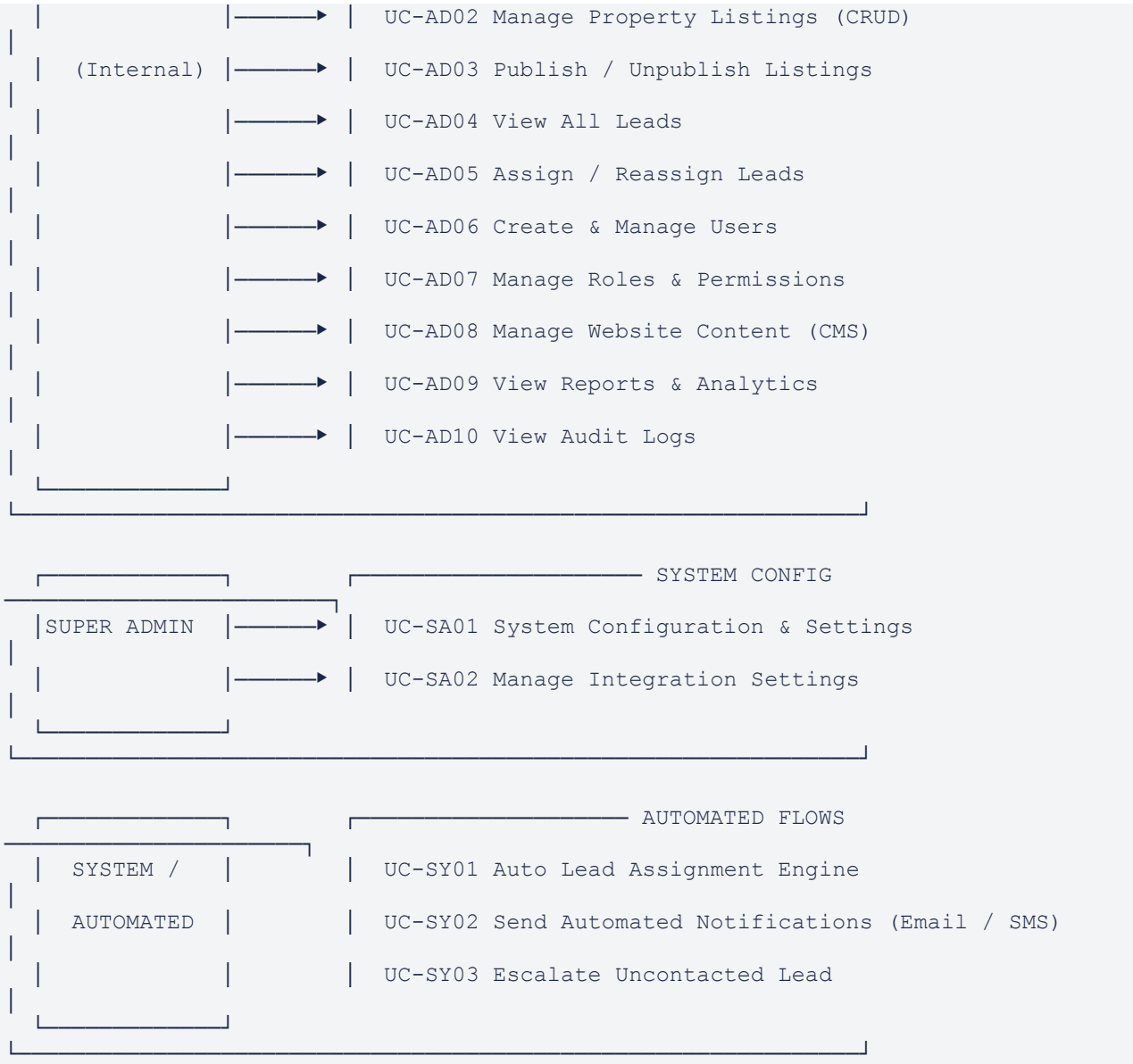
Use Case Count by Actor

Actor	Use Case IDs	Count
Customer	UC-C01 → UC-C08	8
Agent	UC-A01 → UC-A08	8
Admin	UC-AD01 → UC-AD10	10
Super Admin	UC-SA01 → UC-SA02	2
System	UC-SY01 → UC-SY03	3
TOTAL		31

2. USE CASE DIAGRAM — SYSTEM OVERVIEW

The diagram below maps all system actors to their respective use cases across each platform layer.





3. ACTOR DEFINITIONS

Customer

An end-user of the public-facing property website. Customers are unauthenticated visitors or registered users who browse property listings, save favourites, and submit inquiries or leads. They interact exclusively with the public website and their personal account dashboard.

- Motivation: Find a property to buy, rent, or invest in
- Entry Points: Organic search, social media, referral, direct URL, paid ads
- Interaction Level: Self-service; no admin or agent portal access

Agent

An internal sales representative employed by the real estate business. Agents receive leads assigned from the public website, manage client relationships, conduct viewings, and progress deals through the CRM pipeline in the Agent Portal.

- Motivation: Close property deals and manage a pipeline of active leads
- Entry Points: Admin-created account; logs into Agent Portal
- Interaction Level: Agent Portal only; scoped to own leads, customers, and listings

Admin

An internal operations administrator responsible for managing the platform's day-to-day operations. Admins oversee properties, leads, users, website content, and reports. They do not have access to system-level configuration (reserved for Super Admin).

- Motivation: Ensure smooth platform operations, content quality, and team performance
- Entry Points: Admin Portal (admin.domain.com)
- Interaction Level: Admin Portal; full visibility of all agents, leads, and content

Super Admin

The highest-privilege internal user with complete control over the platform. The Super Admin manages role permissions, system integrations, API keys, backup schedules, and global configuration. All other users are created and managed by the Super Admin or delegated to Admin.

- Motivation: Maintain system integrity, security, and configuration
- Entry Points: Admin Portal with elevated credentials and mandatory 2FA
- Interaction Level: All Admin capabilities plus System Configuration module

Email / SMS System (Automated Actor)

The automated notification subsystem triggered by platform events. This is not a human actor — it represents the email service (SendGrid/Postmark) and SMS gateway (Twilio) that send automated communications to Customers, Agents, and Admins based on system rules.

External Portal (Automated Actor)

Third-party property portals (Bayut, Property Finder) that receive property listing data via API sync when a listing is published on the platform. This is an outbound integration actor — the platform pushes data to these systems.

SECTION A — CUSTOMER USE CASES

Use Case ID	UC-C01
Use Case	Register & Create Account
Actor(s)	Customer
Description	A new visitor registers on the public website to create a customer account, enabling them to save properties, track inquiries, and access personalized features.
Pre-conditions	• User has not previously registered with the same email address • Registration page is accessible on the public website
Main Flow	1. Customer clicks 'Register' or 'Sign Up' on the website navigation or inquiry prompt 2. System displays the registration form: Full Name, Email, Phone, Password, Confirm Password 3. Customer fills in all required fields and submits the form 4. System validates input: checks for valid email format, phone format, and password strength 5. System checks that the email address is not already registered 6. System creates a new customer account with status 'Pending Verification' 7. System sends an email verification link to the provided email address 8. Customer clicks the verification link in their email inbox 9. System marks the account as 'Active' and redirects the customer to their dashboard 10. Customer is now logged in and can access all registered-user features
Alt / Exception Flows	1. Email already registered → System displays error: 'An account with this email already exists. Please log in.' 2. Weak password → System highlights password field with strength requirements (min 8 chars, 1 uppercase, 1 number) 3. Customer uses Google / Facebook social login → System creates account using OAuth token; no email verification required 4. Verification link expires (24 hours) → Customer can request a new verification email from the login page 5. Customer closes browser before verifying → Account remains in 'Pending' state; reminder email sent after 24 hours
Post-conditions	• New customer account is created and active in the database • Customer can log in and access their personal dashboard • Admin dashboard 'Registered Customers' counter increments by 1 • Registration event is logged in the system audit trail

Use Case ID	UC-C02
Use Case	Login & Manage Profile
Actor(s)	Customer
Description	A registered customer logs into their account and manages their personal profile, notification preferences, and security settings.
Pre-conditions	• Customer has a verified, active account • Customer knows their email and password
Main Flow	1. Customer clicks 'Login' on the website navigation 2. System displays the login form: Email, Password, Remember Me checkbox 3. Customer enters credentials and clicks 'Login' 4. System validates credentials against the database 5. System creates a session token and logs the customer in

	6. System redirects customer to their account dashboard 7. Customer navigates to 'Edit Profile' to update Name, Phone, or Profile Photo 8. Customer saves changes; system validates and updates the database 9. Customer can also update notification preferences (email, SMS, WhatsApp toggles) 10. Customer can change password by entering current password and a new one
Alt / Exception Flows	1. Incorrect password → System shows error after each attempt; account locked after 5 consecutive failed attempts for 30 minutes 2. Forgot password → Customer clicks 'Forgot Password', enters email, receives reset link valid for 1 hour 3. Account suspended by admin → System shows message: 'Your account has been suspended. Please contact support.' 4. Social login → Customer authenticates via Google/Facebook; session created without password
Post-conditions	• Customer session is active with a valid JWT token • Profile changes are saved and reflected on the account page immediately • Login event is logged with timestamp and IP address

Use Case ID	UC-C03
Use Case	Browse & Search Properties
Actor(s)	Customer
Description	A customer (registered or unregistered) browses the property listings page using search filters and sorting options to find relevant properties.
Pre-conditions	• At least one published property listing exists in the system • Public website is accessible
Main Flow	1. Customer navigates to the 'Properties' section from the main navigation or homepage search bar 2. System displays all published listings in default order (newest first) with pagination 3. Customer applies filters: Property Type, Status (Sale/Rent), Location, Price Range, Bedrooms, Bathrooms, Area, Furnishing, Amenities 4. System dynamically filters and refreshes the results list with matching listings 5. Customer applies a sort option: Price (Low–High), Price (High–Low), Newest, Most Viewed 6. System re-orders results according to the selected sort 7. Customer toggles between Grid View, List View, and Map View 8. On Map View, customer sees property pin markers on an interactive map and can click pins to preview cards 9. Customer clicks on a property card to view the full property detail page
Alt / Exception Flows	1. No results match filters → System shows 'No properties found matching your criteria' with a 'Clear Filters' button 2. Customer searches by keyword (building name, area) → System performs text search across title, description, and location fields 3. Customer is on mobile → Filters appear in a slide-up drawer for responsive experience
Post-conditions	• Customer has a filtered, relevant list of properties matching their requirements • Applied filter state is preserved in the URL (shareable/bookmarkable links) • Property view counts are incremented for analytics tracking

Use Case ID	UC-C04
Use Case	View Property Details

Actor(s)	Customer
Description	A customer views the full detail page of a specific property listing including media, specifications, location, and agent information.
Pre-conditions	• The property is published and active • Customer has access to the public website (logged in or anonymous)
Main Flow	1. Customer clicks on a property card from the listings page or a search result 2. System loads the Property Detail Page for the selected listing 3. System records a view event for this property (used in analytics) 4. Customer views the image gallery and optionally plays a video tour or virtual tour 5. Customer reads the property overview, description, and specifications 6. Customer navigates tabs: Overview, Details, Amenities, Location, Payment Plan, Floor Plan 7. Customer views the embedded map and explores nearby points of interest 8. Customer views the assigned agent's card: name, photo, phone, WhatsApp, email 9. Customer scrolls to the Inquiry Form and considers submitting a lead 10. Customer views 'Similar Properties' carousel at the bottom of the page
Alt / Exception Flows	1. Property has been unpublished or archived → System shows '404 – Property Not Found' page with a link to browse listings 2. Customer clicks agent's WhatsApp link → Opens WhatsApp with pre-filled message including property reference 3. Customer clicks 'Share' → Social sharing options appear (WhatsApp, Email, Facebook, copy link) 4. Customer not logged in and tries to save → Redirected to login/register page with return URL preserved
Post-conditions	• Property view count is incremented in the analytics system • Customer has full information to make an inquiry decision • Property reference number is pre-filled in inquiry form if customer proceeds

Use Case ID	UC-C05
Use Case	Save Property to Wishlist
Actor(s)	Customer
Description	A registered customer saves a property listing to their wishlist/favourites for future reference.
Pre-conditions	• Customer is logged in with an active account • The target property is published
Main Flow	1. Customer views a property (from listing card or detail page) 2. Customer clicks the heart/bookmark icon on the property card or detail page 3. System checks that the customer is logged in 4. System adds the property to the customer's saved list in the database 5. Icon changes to a filled/active state to indicate it has been saved 6. Customer navigates to 'My Account → Saved Properties' to view their wishlist 7. Customer can remove a saved property by clicking the icon again or from the wishlist page
Alt / Exception Flows	1. Customer not logged in → Clicking save icon redirects to login with return URL, property is saved after authentication 2. Property already saved → Clicking icon removes it from wishlist (toggle behaviour) 3. Saved property is later archived or unpublished → It remains in wishlist but is marked 'No Longer Available'

Post-conditions	- Property is stored in customer's wishlist in the database - Wishlist is visible in customer's account dashboard - Wishlist data can be used by agents to understand customer interest (visible in lead notes if customer submits an inquiry)
Use Case ID	UC-C06
Use Case	Submit Property Inquiry / Lead
Actor(s)	Customer
Description	A customer submits an inquiry about a specific property through the inquiry form on the property detail page. This creates a lead record in the system and triggers the auto-assignment engine.
Pre-conditions	- The property is published and active - Customer has internet access (login not required)
Main Flow	- Customer is on the Property Detail Page and decides to inquire - Customer fills in the Inquiry Form: Name, Email, Phone Number, Message (optional) - Property reference number is pre-filled by the system - Customer checks the 'I agree to the privacy policy' checkbox - Customer clicks 'Send Inquiry' - System validates all fields: required fields, valid email format, valid phone format - System creates a new Lead record with status 'New', linked to the property and customer details - System triggers UC-SY01 (Auto Lead Assignment Engine) - System triggers UC-SY02 (Send notification: confirmation email to customer, alert to admin) - Customer sees a success message: 'Thank you! An agent will contact you within 24 hours.' - If customer is logged in, the inquiry appears in their 'My Inquiries' section
Alt / Exception Flows	- Required field missing → System highlights the empty field with an error message; form is not submitted - Invalid email format → System shows inline validation error on the email field - Duplicate inquiry (same email + same property within 24 hours) → System shows message: 'You have already submitted an inquiry for this property. Our agent will be in touch soon.' - Spam / bot detection fails (captcha) → Form submission blocked with error message - Customer not logged in → Lead is created with provided contact details; customer is prompted (not forced) to register
Post-conditions	- A Lead record is created in the database with status 'New' - Lead is linked to the property, customer contact details, and source URL - Auto Lead Assignment Engine runs (UC-SY01) - Confirmation email is sent to customer (UC-SY02) - New Lead alert is sent to admin (UC-SY02) - Lead appears in Admin Portal → All Leads table
Use Case ID	UC-C07
Use Case	Contact an Agent Directly
Actor(s)	Customer
Description	A customer contacts a specific agent via the Agent Directory or an Agent Profile Page, creating a lead tagged to that agent.

Pre-conditions	● Agent profile is published and active ● Customer is on the Agents Directory or an Agent Profile Page
Main Flow	1. Customer navigates to the Agents Directory from the website navigation 2. Customer browses agent profiles or searches by name, specialization, or language 3. Customer clicks on an agent card to open the Agent Profile Page 4. Customer reviews the agent's bio, listings, specialization, and reviews 5. Customer fills in the contact form on the agent profile: Name, Email, Phone, Message 6. Customer clicks 'Contact Agent' 7. System creates a Lead record linked to that specific agent (bypasses auto-assignment for the agent field) 8. System sends a notification to the agent and admin 9. Customer receives confirmation message
Alt / Exception Flows	1. Customer clicks 'Call' button → Opens phone dialer with agent's number (no lead created automatically; admin can manually log) 2. Customer clicks 'WhatsApp' button → Opens WhatsApp with agent's number and pre-filled message 3. Agent is currently inactive/unavailable → Contact form is still shown; lead goes to admin for manual reassignment
Post-conditions	● Lead is created and directly assigned to the selected agent ● Agent receives immediate notification ● Admin can view and override the assignment if needed

Use Case ID	UC-C08
Use Case	Track Inquiry / Lead Status
Actor(s)	Customer
Description	A logged-in customer views the status and progress of their submitted inquiries from their personal account dashboard.
Pre-conditions	● Customer has a registered, verified, and active account ● Customer has at least one submitted inquiry
Main Flow	1. Customer logs into their account 2. Customer navigates to 'My Account → My Inquiries' 3. System displays a list of all inquiries submitted by the customer: Property, Date, Agent Assigned, Current Status 4. Customer clicks on a specific inquiry to view its detail 5. System shows: property details (thumbnail, title), assigned agent contact info, current status, last update date 6. Customer can resend or follow up via the contact form on the same page
Alt / Exception Flows	1. No inquiries submitted yet → System shows empty state: 'You haven't submitted any inquiries yet' with a CTA to browse properties 2. Property has been archived since the inquiry → Property shown with 'No Longer Available' badge; inquiry record remains
Post-conditions	● Customer has visibility into their inquiry pipeline ● Customer knows the assigned agent and can initiate contact

SECTION B — AGENT USE CASES

Use Case ID	UC-A01
Use Case	Login to Agent Portal
Actor(s)	Agent
Description	An agent authenticates into the Agent Portal using credentials created by the Admin.
Pre-conditions	• Agent account has been created by Admin with 'Agent' role • Account status is 'Active' • Agent has received login credentials (email + temporary password) via welcome email
Main Flow	1. Agent navigates to the Agent Portal URL (portal.domain.com) 2. System displays the login page with Email and Password fields 3. Agent enters credentials and clicks 'Login' 4. System validates credentials and role (must be Agent or Admin) 5. System creates a scoped session token limited to agent's own data 6. System redirects agent to the Agent Dashboard 7. Agent views their dashboard: new leads, today's tasks, upcoming viewings, performance KPIs
Alt / Exception Flows	1. First-time login → System prompts agent to set a new password (temporary password must be changed) 2. Incorrect credentials → Error shown; account locked after 5 failed attempts 3. Session expires → Agent redirected to login page with message: 'Your session has expired. Please log in again.' 4. Admin has suspended agent account → Login blocked with message directing to contact admin
Post-conditions	• Agent is authenticated with an active scoped session • Login event is recorded in the audit trail with timestamp and IP address • Dashboard loads with real-time lead and task data

Use Case ID	UC-A02
Use Case	View & Manage Assigned Leads
Actor(s)	Agent
Description	An agent views their full list of assigned leads, filters and sorts them, and accesses individual lead detail pages for follow-up and management.
Pre-conditions	• Agent is logged in • At least one lead has been assigned to the agent
Main Flow	1. Agent navigates to 'My Leads' in the Agent Portal 2. System displays a table of all leads assigned to this agent: Customer Name, Phone, Property, Status, Date Assigned, Priority 3. Agent filters leads by Status (New, Active, Viewing, etc.) or by Priority flag 4. Agent searches for a specific customer by name, phone, or email 5. Agent clicks a lead row to open the Lead Detail Page 6. Lead Detail Page shows: Customer info, Property info, Lead status pipeline, Activity timeline, Action buttons 7. Agent reviews the customer's details, property interest, and all previous interactions 8. Agent decides on the next action (call, email, note, update status)

Alt / Exception Flows	1. Agent has no leads assigned → Dashboard shows empty state with message: 'No leads assigned yet.' 2. Lead was reassigned away from agent by Admin → Lead disappears from agent's list; agent receives notification 3. New lead assigned while agent is logged in → Real-time notification appears in the portal notification bell
Post-conditions	• Agent has a clear, prioritized view of their pipeline • Agent can act on any lead from the detail page

Use Case ID	UC-A03
Use Case	Update Lead Status & Progress
Actor(s)	Agent
Description	An agent updates the status of a lead to reflect the current stage in the sales pipeline, providing real-time visibility to the Admin.
Pre-conditions	• Agent is logged in and on a Lead Detail Page • The lead is assigned to this agent
Main Flow	1. Agent reviews current lead status in the pipeline bar 2. Agent clicks the next status stage button or selects from the status dropdown 3. System validates the status transition (e.g., cannot skip from 'New' to 'Won' without intermediate steps — configurable rule) 4. System saves the new status with a timestamp 5. System automatically adds an entry to the Activity Timeline: 'Status updated from X to Y by [Agent Name] on [Date/Time]' 6. System triggers UC-SY02 if the new status is 'Won' or requires customer notification 7. Admin's portal updates in real time to reflect the new status 8. If status = 'Lost', agent is required to select a Loss Reason from a dropdown (e.g., Price too high, Found another property, Unresponsive, Budget changed) 9. If status = 'Won', agent enters the deal value and expected closing date
Alt / Exception Flows	1. Agent marks as 'Lost' without selecting a reason → System blocks submission with validation error: 'Please select a reason for loss' 2. Admin overrides status → Admin's change is recorded separately in the timeline with admin's name 3. Agent accidentally sets wrong status → Admin can manually correct the status; correction is logged
Post-conditions	• Lead status is updated in real time in both Agent and Admin portals • Activity timeline entry is created with timestamp and actor • Admin report data is updated to reflect the new pipeline stage • If Won: conversion is counted in the agent's performance metrics • If Lost: loss reason is recorded for funnel analysis reporting

Use Case ID	UC-A04
Use Case	Log Call & Communication Activity
Actor(s)	Agent
Description	An agent logs a phone call or other communication activity against a lead, maintaining a complete interaction history.
Pre-conditions	• Agent is logged in and on a Lead Detail Page • The lead is assigned to this agent

Main Flow	1. Agent clicks 'Log Call' on the Lead Detail Page 2. System opens a Log Call form: Date/Time (pre-filled with current), Duration (minutes), Outcome dropdown, Notes, Next Follow-up Date 3. Outcome options: Reached – Positive, Reached – Neutral, Reached – Not Interested, Not Answered, Voicemail Left, Number Wrong 4. Agent fills in the form and clicks 'Save' 5. System creates an Activity Timeline entry: 'Call logged by [Agent] on [Date]: [Outcome] — [Notes]' 6. If a follow-up date was set, system creates a reminder that appears in agent's Today's Tasks on that date 7. Agent can also use 'Add Note' for non-call communications (WhatsApp, email summaries)
Alt / Exception Flows	1. Agent logs email sent from external email client → Uses 'Add Note' with type 'Email' to manually record it 2. Customer calls agent proactively → Agent still logs the call for the record 3. Follow-up reminder date passes without action → Reminder remains in Today's Tasks highlighted in amber as overdue
Post-conditions	• Call is recorded in the lead's Activity Timeline • Follow-up reminder is scheduled if entered • Admin can view the complete call log from the lead detail in the Admin Portal • Time-to-contact metric is tracked from lead assignment to first logged call

Use Case ID	UC-A05
Use Case	Schedule Property Viewing
Actor(s)	Agent
Description	An agent schedules a property viewing appointment with a customer, updates the lead status, and adds the viewing to the Agent Calendar.
Pre-conditions	• Agent is on a Lead Detail Page • Lead status is at least 'Contacted' or 'Active' • A property is linked to the lead
Main Flow	1. Agent clicks 'Schedule Viewing' on the Lead Detail Page 2. System opens a Viewing Scheduler form: Date, Time, Property (pre-filled from lead), Meeting Location (on-site / off-plan showroom / virtual), Notes 3. Agent fills in the viewing details and clicks 'Schedule' 4. System saves the viewing appointment 5. System updates the lead status to 'Viewing Scheduled' 6. System adds an Activity Timeline entry: 'Viewing scheduled for [Date/Time] at [Location]' 7. System adds the event to the Agent Calendar 8. System triggers UC-SY02: sends viewing confirmation to the customer (Email + SMS with date, time, and location) 9. Agent receives a calendar reminder notification 1 hour before the viewing
Alt / Exception Flows	1. Customer does not confirm → Agent can mark viewing as 'Pending Confirmation' and follow up 2. Viewing needs to be rescheduled → Agent edits the viewing entry; rescheduling notification sent to customer 3. Viewing is cancelled → Agent marks as cancelled; lead reverts to 'Active' status; cancellation reason noted 4. Google Calendar sync enabled → Viewing event is pushed to agent's Google Calendar automatically

Post-conditions	- Viewing appointment is saved in the system - Customer has received a confirmation communication - Lead status is updated to 'Viewing Scheduled' - Viewing appears in agent's Calendar module - Admin can see upcoming viewings in the lead detail and in overall activity
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Use Case ID	UC-A06
Use Case	Manage Customer Profile & Notes
Actor(s)	Agent
Description	An agent views and updates customer profile information and requirements to support better lead management and personalised property recommendations.
Pre-conditions	- Agent is logged in - Customer is linked to one of the agent's leads
Main Flow	- Agent navigates to 'My Customers' in the Agent Portal - System displays a list of all unique customers linked to the agent's leads - Agent clicks on a customer to open their profile - System shows: contact details, all linked leads (with statuses), full interaction history across all leads - Agent edits customer requirements notes: budget range, preferred areas, property types, timeline, family size - Agent updates contact preferences: best time to call, preferred channel - Agent saves changes; system updates the customer profile - Changes are visible in all lead detail pages linked to this customer
Alt / Exception Flows	- Customer's phone or email has changed → Agent updates contact details; change is logged in the audit trail - Customer has multiple active leads → Agent sees all leads on the customer profile and can navigate to any one
Post-conditions	- Customer profile is updated with accurate requirements and contact preferences - Updated notes inform future property recommendations by the agent - Admin can view the updated customer profile from the Admin Portal

Use Case ID	UC-A07
Use Case	Manage Property Listings
Actor(s)	Agent
Description	An agent manages property listings assigned to them — updating details, uploading new media, and flagging status changes for admin review.
Pre-conditions	- Agent is logged in - Listings have been assigned to the agent by Admin
Main Flow	- Agent navigates to 'My Properties' in the Agent Portal - System displays all listings assigned to this agent with: thumbnail, title, status, inquiry count, last updated date - Agent clicks a listing to open its edit form - Agent updates allowed fields (configurable by Admin): description, price, availability, media photos - Agent uploads new property photos or updates the gallery order - Agent clicks 'Save Changes'

	7. System saves the changes (if property is already Published, changes are live immediately or go to draft pending admin review — based on admin setting) 8. Agent can mark a property as 'Under Offer' — triggers an admin notification for review 9. Agent can flag a property as 'Sold' or 'Rented' — triggers admin review before archiving
Alt / Exception Flows	1. Agent tries to delete a listing → Action not permitted; agent sees message: 'Contact admin to remove a listing' 2. Agent tries to publish/unpublish → Action not permitted; agent sees message: 'Publication is managed by the admin team' 3. Admin has restricted editing to certain fields only → Other fields appear as read-only in the edit form
Post-conditions	- Property details are updated in the system - If published, changes are reflected on the public website immediately (or after admin approval) - Update event is logged in the audit trail

Use Case ID	UC-A08
Use Case	View Personal Performance Dashboard
Actor(s)	Agent
Description	An agent views their personal performance metrics and KPIs for the current period on the Agent Dashboard.
Pre-conditions	- Agent is logged in - Agent has at least some lead activity in the system
Main Flow	1. Agent views the Performance section on their dashboard or navigates to 'My Performance' 2. System calculates and displays: Total leads assigned this month vs. last month, New leads today, Leads contacted within 1 hour / 24 hours, Conversion rate (Lead → Won), Number of viewings conducted, Deals closed and total value 3. System displays trend arrows (up/down) comparing current month to previous month 4. Agent views bar/line charts for monthly lead trends 5. Agent can filter by date range (This Week, This Month, Last Month, This Quarter)
Alt / Exception Flows	1. No data for selected period → Charts show empty state with message: 'No activity recorded for this period' 2. Admin has restricted performance visibility → Certain metrics (e.g., deal values) may be hidden based on role config
Post-conditions	- Agent has visibility into their own KPIs - Agent can self-monitor and adjust activity based on metrics - No data is modified — this is a read-only view

SECTION C — ADMIN USE CASES

Use Case ID	UC-AD01
Use Case	Login to Admin Portal
Actor(s)	Admin / Super Admin
Description	An Admin or Super Admin authenticates into the Admin Portal with optional Two-Factor Authentication.
Pre-conditions	- Admin account exists with 'Admin' or 'Super Admin' role - Account status is 'Active' 2FA is configured (mandatory for Super Admin; optional for Admin)
Main Flow	- Admin navigates to the Admin Portal URL (admin.domain.com) - System displays the admin login page with Email and Password fields - Admin enters credentials and clicks 'Login' - System validates credentials and confirms the role is Admin or Super Admin - If 2FA is enabled: System prompts for a 6-digit TOTP code from the authenticator app - Admin enters the 2FA code; system validates it - System creates an admin session with appropriate role-scoped permissions - System redirects admin to the Admin Dashboard - Dashboard loads with real-time KPI cards, activity feed, and recent leads
Alt / Exception Flows	- Invalid 2FA code → Error shown; admin can retry up to 3 times before being locked out - Admin forgets 2FA device → Super Admin can disable 2FA for the account temporarily; re-setup required on next login - IP not in allowlist (if IP restriction is configured) → Login blocked regardless of credentials
Post-conditions	- Admin is authenticated with a role-scoped session token - Login is recorded in the Audit Log: user, timestamp, IP address, success/failure

Use Case ID	UC-AD02
Use Case	Manage Property Listings (Full CRUD)
Actor(s)	Admin
Description	An Admin creates, reads, updates, and deletes property listings — managing the complete lifecycle of all listings on the platform.
Pre-conditions	Admin is logged in with Property Management permissions
Main Flow	- Admin navigates to 'Properties' in the Admin Portal - System displays the listings table with search, filter, and bulk-action tools - To Add: Admin clicks 'Add Property', fills in all sections of the property form (Basic Info, Location, Details, Media, SEO, Agent Assignment) and saves - System creates the listing with status 'Draft' - To Edit: Admin clicks the edit icon on any listing row, modifies fields, and saves - System updates the listing; if already Published, changes apply live to the website immediately - To Delete: Admin clicks delete, confirms in the dialog box - System soft-deletes the listing (archived, not permanently erased) unless hard-delete is explicitly chosen - To Bulk Action: Admin selects multiple listings via checkboxes, chooses a bulk action (Publish, Unpublish, Archive, Reassign Agent) - System applies the action to all selected listings and logs each change

Alt / Exception Flows	1. Required fields missing during add/edit → System highlights empty required fields; save is blocked 2. Duplicate reference number → System shows error: 'A listing with this reference number already exists' 3. Media upload fails (file too large or wrong format) → Error displayed; admin uploads a compliant file 4. Attempting to delete a listing with active leads → System warns: 'This listing has X active leads. Deleting will not remove the leads.' Admin must confirm
Post-conditions	• Listing record is created, updated, or archived in the database • All changes are recorded in the Audit Log with before/after field values • Published listings are immediately visible on the public website • Deleted/archived listings are removed from public website but remain in admin for record

Use Case ID	UC-AD03
Use Case	Publish / Unpublish Listings
Actor(s)	Admin
Description	An Admin controls the visibility of property listings on the public website by publishing or unpublishing them.
Pre-conditions	• Admin is logged in • Listing exists in Draft, Published, or Pending Review status
Main Flow	1. Admin locates the listing in the Properties table 2. Admin clicks the status toggle or selects 'Publish' from the actions menu 3. System validates that all required fields are complete (title, price, location, at least one photo) 4. System changes status to 'Published' and sets the publication timestamp 5. Listing becomes visible on the public website immediately 6. For unpublishing: Admin selects 'Unpublish' — listing status changes to 'Draft'; listing disappears from the website 7. System logs the status change in the Audit Trail 8. If portal sync is configured: System sends the listing to Bayut/Property Finder via API (on publish) or withdraws it (on unpublish)
Alt / Exception Flows	1. Listing missing required fields → Publish blocked; system shows a checklist of missing items 2. Scheduled publish: Admin sets a future publish date → System publishes automatically at the scheduled date/time 3. Agent submitted a listing for review → Admin reviews and approves (publishes) or rejects (sends back to Draft with notes)
Post-conditions	• Listing status is updated; website reflects the change in real time • External portal sync is triggered (if enabled) • Publish/unpublish event is logged in Audit Trail

Use Case ID	UC-AD04
Use Case	View & Manage All Leads
Actor(s)	Admin
Description	An Admin views the complete leads database across all agents, monitors pipeline health, and manages individual lead records.
Pre-conditions	• Admin is logged in

	At least one lead exists in the system
Main Flow	- Admin navigates to 'Leads' in the Admin Portal - System displays all leads across all agents in a table: Lead ID, Customer, Property, Source, Agent, Status, Date, Last Activity - Admin applies filters: Status, Agent, Date Range, Lead Source, Property Type - Admin searches for a specific lead by customer name, phone, or email - Admin clicks a lead row to open the Lead Detail page - Admin reviews full customer info, property interest, activity timeline, and agent notes - Admin can add Admin Notes (visible only to admin, not agent or customer) - Admin can manually override the lead status - Admin can reassign the lead to a different agent (triggers UC-AD05)
Alt / Exception Flows	- Admin uses bulk actions to archive stale leads (e.g., no activity in 90 days) → System asks for confirmation; bulk archive is logged - Admin exports leads to Excel → System generates and downloads a CSV with all visible columns - Admin filters to 'Unassigned' leads → Sees all new leads not yet processed by auto-assignment for manual handling
Post-conditions	- Admin has complete visibility into all lead activity across the platform - Any changes (status, reassignment, notes) are saved and logged - Lead data informs reports and agent performance tracking

Use Case ID	UC-AD05
Use Case	Assign / Reassign Leads to Agents
Actor(s)	Admin
Description	An Admin manually assigns new leads or reassigns existing leads to different agents based on workload, expertise, or business need.
Pre-conditions	- Admin is logged in - At least one lead exists (unassigned or assigned) - At least one active agent exists
Main Flow	- Admin opens a Lead Detail page (from Leads table or Dashboard new leads widget) - Admin clicks 'Assign Agent' (for new leads) or 'Reassign Agent' (for existing leads) - System displays a list of available agents with: name, current lead count, specialization, availability status - Admin selects the target agent from the list - Admin optionally adds a note explaining the assignment/reassignment reason - Admin confirms the action - System assigns the lead to the selected agent - System triggers UC-SY02: new agent receives Email + SMS + Portal notification - Activity Timeline entry is created: 'Lead assigned to [Agent] by [Admin] on [Date/Time]' - If reassignment: previous agent is notified that the lead has been moved from their pipeline
Alt / Exception Flows	- Selected agent has reached their lead cap → System warns: 'This agent has X leads (cap: Y). Do you still want to assign?' Admin can override - No agents available → Admin sees message with option to create a new agent account - Bulk reassignment → Admin selects multiple leads from the table and reassigns all to one agent in one action
Post-conditions	- Lead is now assigned (or reassigned) to the selected agent - Both old and new agents (if applicable) have been notified - Assignment is recorded in the Audit Trail

	● Lead appears in the new agent's portal immediately
Use Case ID	UC-AD06
Use Case	Create & Manage Users
Actor(s)	Admin
Description	An Admin creates new user accounts for Agents, Admins, and Customers, and manages existing user records including status and profile details.
Pre-conditions	● Admin is logged in with User Management permissions
Main Flow	1. Admin navigates to 'Users' in the Admin Portal 2. System displays all users across all roles in a filterable, searchable table 3. To Create: Admin clicks 'Add User', selects the role (Admin/Agent/Customer), fills in Name, Email, Phone, Password (or auto-generate), and optionally sets Agent-specific fields (bio, languages, specialization, photo) 4. Admin clicks 'Create User' 5. System creates the user account and sends a welcome email with credentials 6. To Edit: Admin clicks the edit icon, updates fields, saves 7. To Deactivate: Admin toggles account status to 'Inactive' or 'Suspended' — user can no longer log in 8. To Delete: Super Admin only can permanently delete a user after confirming there are no active records linked
Alt / Exception Flows	1. Email already in use → System blocks creation and shows error 2. Auto-generate password selected → System creates a secure random password, includes it in the welcome email, and forces a password reset on first login 3. Admin tries to delete a user with active leads → System warns: 'This agent has X active leads. Reassign leads before deletion.' 4. Creating a Customer account manually → Useful when a customer inquired by phone and the admin creates their profile
Post-conditions	● User account is created and active in the database ● Welcome email is sent to the new user with login credentials ● User creation is logged in the Audit Trail with admin's name ● For agents: profile appears on the Agent Directory on the public website

Use Case ID	UC-AD07
Use Case	Manage Roles & Permissions
Actor(s)	Admin
Description	An Admin (with permission) or Super Admin manages role definitions and the permission matrix controlling what each role can access and do within the platform.
Pre-conditions	● Admin or Super Admin is logged in ● Role Management permission is granted
Main Flow	1. Admin navigates to 'Settings → Roles & Permissions' 2. System displays the permission matrix: rows = features/modules, columns = roles 3. Admin reviews current permissions for each role 4. Admin clicks on a cell to toggle a permission (grant or revoke) 5. System shows a confirmation prompt: 'This will affect all users with the [Role] role. Confirm?' 6. Admin confirms; system saves the updated permissions 7. All users with that role immediately have their access updated on next page load / session refresh

	8. Admin can also create a new custom role (e.g., 'Marketing Admin') and define its permissions from scratch 9. Admin assigns the new role to selected users
Alt / Exception Flows	1. Admin tries to revoke a permission from Super Admin role → Action blocked; Super Admin permissions are locked 2. Admin creates a role with no permissions → System warns but allows save (empty role) 3. Saving conflicts detected (e.g., granting 'Delete Users' to Admin when Super Admin restriction is set) → System blocks with explanation
Post-conditions	- Updated permissions take effect immediately - Permission change is logged in Audit Trail: 'Role [X] permission [Y] changed from [Granted/Denied] by [Admin]' - All users of the modified role experience the updated access on next interaction

Use Case ID	UC-AD08
Use Case	Manage Website Content (CMS)
Actor(s)	Admin
Description	An Admin manages all dynamic content displayed on the public website — including banners, pages, blog articles, testimonials, navigation, and SEO settings.
Pre-conditions	- Admin is logged in with CMS permissions - Public website is connected to the CMS backend
Main Flow	1. Admin navigates to 'Content' in the Admin Portal 2. Admin selects the content type to manage: Homepage, Pages, Blog, Navigation, Amenities, Announcements 3. For Homepage: Admin adds/edits/deletes hero banners (image upload, headline, CTA link, order, active toggle) 4. Admin selects which listings appear as Featured Properties on the homepage 5. Admin manages testimonials: adds new reviews, edits existing, toggles visibility 6. For Pages: Admin opens a static page (e.g., About Us) in the WYSIWYG editor, makes changes, saves 7. For Blog: Admin creates a new article with title, body (rich text), featured image, category, tags, author, SEO fields, publish date 8. Admin saves content; published changes are immediately reflected on the public website 9. Admin can schedule content to go live at a future date/time
Alt / Exception Flows	1. Admin saves a page without SEO fields → System shows warning: 'SEO meta title and description are empty. This may affect search rankings.' Admin can save anyway 2. Banner image exceeds file size limit → System rejects upload with message and size limit 3. Blog article is saved as Draft (not published) → Appears only in admin CMS, not on public website
Post-conditions	- Published content changes are live on the public website - All CMS edits are logged in the Audit Trail with the admin's name and a before/after snapshot - SEO metadata is updated for search engine indexing

Use Case ID	UC-AD09
Use Case	View Reports & Analytics
Actor(s)	Admin

Description	An Admin generates, views, and exports reports covering lead performance, agent activity, property metrics, and business conversion analytics.
Pre-conditions	- Admin is logged in with Reports access - Sufficient data exists in the system for the selected date range
Main Flow	- Admin navigates to 'Reports' in the Admin Portal - System displays available report types: Lead Summary, Agent Performance, Property Performance, Conversion Funnel, Customer Activity, Revenue/Deals, Source Attribution - Admin selects a report type and sets the date range (custom or preset: Today/Week/Month/Quarter/Year) - System generates the report with charts and data tables - Admin reviews: KPI numbers, trend charts (bar, line, pie, funnel), data breakdowns - Admin applies additional filters (e.g., by agent, by property type, by area) - Admin clicks 'Export' and selects format (PDF or Excel/CSV) - System generates and downloads the report file - Admin can set up scheduled reports: select report type, frequency (daily/weekly/monthly), and recipient email addresses
Alt / Exception Flows	- No data for selected range → Report shows empty state charts with message: 'No data available for this period' - Scheduled report fails to send → System retries 3 times; logs failure in audit trail; admin is notified via portal alert - Admin wants a custom report not pre-built → Custom report builder (Phase 2) allows field selection and custom filters
Post-conditions	- Admin has actionable business insights - Exported report files are available for sharing or archiving - Scheduled reports continue to run without further admin input - Report generation is logged in Audit Trail

Use Case ID	UC-AD10
Use Case	View Audit Logs
Actor(s)	Super Admin
Description	The Super Admin views a comprehensive, tamper-proof log of all system actions for compliance, security review, and accountability.
Pre-conditions	- Super Admin is logged in - Audit Log module is available (Super Admin access only)
Main Flow	- Super Admin navigates to 'Settings → Audit Logs' - System displays a chronological log of all recorded events - Each log entry shows: Timestamp, Actor (user name + role), Action Type, Module, Details (before → after values), IP Address, Session ID - Super Admin filters by: User, Action Type (Login/Property/Lead/User/CMS/Config), Module, Date Range - Super Admin searches for a specific event by keyword or user name - Super Admin clicks an entry to view full detail of the event - Super Admin exports the filtered log to CSV for compliance audit or external review
Alt / Exception Flows	- Super Admin tries to edit or delete an audit entry → Action blocked: 'Audit logs are read-only and cannot be modified.' - Log volume is very large → System paginates; Super Admin can filter to narrow scope before exporting
Post-conditions	- Super Admin has a complete, verified record of platform activity - Exported log is available for external compliance or security review

	No data is modified — Audit Log is strictly read-only
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SECTION D — SUPER ADMIN USE CASES

Use Case ID	UC-SA01
Use Case	System Configuration & Settings
Actor(s)	Super Admin
Description	The Super Admin configures global system settings including general platform details, email/SMS gateways, SEO settings, analytics integrations, and backup schedules.
Pre-conditions	• Super Admin is logged in with 2FA verified • System Configuration module is accessible
Main Flow	1. Super Admin navigates to 'Settings → System Configuration' 2. System displays configuration categories: General, Email & Notifications, Lead Assignment, SEO & Analytics, Backup & Maintenance 3. General Settings: Super Admin updates website name, logo, favicon, contact details, default currency, measurement unit, time zone; saves 4. Email Settings: Super Admin configures SMTP credentials (host, port, username, password, sender name); tests the connection with a test email 5. SMS Settings: Super Admin enters Twilio API credentials (Account SID, Auth Token, From Number); tests with a test SMS 6. Lead Assignment Settings: Super Admin configures auto-assignment method (Round-Robin / Area-based / Manual), sets per-agent lead cap, sets uncontacted lead escalation time (default: 60 minutes) 7. SEO & Analytics: Super Admin enters Google Analytics ID, GTM ID, Facebook Pixel ID; toggles sitemap auto-generation; edits robots.txt 8. Backup Settings: Super Admin sets backup frequency and retention period; triggers a manual backup 9. Maintenance Mode: Super Admin toggles maintenance mode ON/OFF with a custom maintenance message 10. All changes are saved and applied immediately; critical changes (email, SMS) are tested before save is confirmed
Alt / Exception Flows	1. Email test fails → System shows SMTP error details; Super Admin corrects credentials and retests 2. Super Admin sets maintenance mode ON → All public website visitors see the maintenance page; admin portal remains accessible 3. Backup fails → System retries; logs failure; sends alert to Super Admin email
Post-conditions	• System configuration is updated and applied globally • All configuration changes are logged in the Audit Trail • Email and SMS services are operational with new credentials • Public website reflects general setting changes (logo, name, etc.) immediately

Use Case ID	UC-SA02
Use Case	Manage Integration Settings
Actor(s)	Super Admin
Description	The Super Admin configures third-party integrations including property portal syndication (Bayut, Property Finder), CRM systems, and payment gateways.
Pre-conditions	• Super Admin is logged in • API credentials for the target integrations are available
Main Flow	1. Super Admin navigates to 'Settings → Integrations'

	2. System displays available integrations with connection status (Connected / Not Connected) 3. Property Portal Sync: Super Admin enters API key and broker ID for Bayut and/or Property Finder; enables sync toggle; sets sync frequency (real-time / hourly / daily) 4. System performs a test sync with a sample listing and confirms connection 5. CRM Integration (optional): Super Admin enters CRM API endpoint and authentication token (for Salesforce/HubSpot); maps platform lead fields to CRM fields 6. Payment Gateway: Super Admin enters gateway API keys (for featured listing payments); tests a payment in sandbox mode 7. WhatsApp Business API: Super Admin enters WABA credentials and configures message templates for automated notifications 8. Super Admin saves all integration settings 9. System validates each integration connection and shows green/red status indicators
Alt / Exception Flows	1. Integration API key is invalid → Connection test fails; Super Admin sees error message with API provider details 2. Portal sync fails for a specific listing → Listing is flagged in the Properties table; Super Admin or Admin can retry sync manually 3. CRM field mapping incomplete → System warns that unmapped required fields will be left empty in CRM; Super Admin confirms
Post-conditions	● Integrations are active and operational ● Property listings are automatically synced to enabled external portals on publish ● Leads created via external portals (if supported) are imported into the platform ● Integration setup is logged in the Audit Trail

SECTION E — SYSTEM / AUTOMATED USE CASES

Use Case ID	UC-SY01
Use Case	Auto Lead Assignment Engine
Actor(s)	System (Automated — triggered by UC-C06)
Description	The system automatically assigns a newly created lead to the most appropriate agent based on a configurable 4-rule cascading logic, without any manual admin intervention.
Pre-conditions	• A new Lead record has been created (status: New) • At least one active Agent account exists in the system • Auto-assignment is enabled in System Configuration (UC-SA01)
Main Flow	1. System receives trigger: new Lead record created 2. Rule 1 — Property Agent Check: System checks if the property linked to the lead has an assigned agent. If YES → assign lead to that agent, go to Step 8 3. Rule 2 — Area Specialization Check: System reads the property's community/area and checks if any active agent has that area listed as a specialization. If YES → among matching agents, select the one with the fewest active leads (lowest load). Assign and go to Step 8 4. Rule 3 — Round-Robin Queue: System selects the next agent in the round-robin rotation queue. System checks if that agent's current lead count is below their configured lead cap. If below cap → assign and go to Step 8. If at cap → skip to next agent in queue. If all agents are at cap → proceed to Rule 4 5. Rule 4 — Admin Fallback: System assigns the lead to the Admin/Team Leader account and flags it as 'Requires Manual Assignment'. Admin receives an immediate high-priority alert 6. Assignment: System updates the Lead record — sets Agent ID, Assignment Timestamp, Status → 'Assigned' 7. System triggers UC-SY02: notifies the assigned agent (Email + SMS + Portal notification) 8. System logs the assignment in the Activity Timeline and Audit Trail
Alt / Exception Flows	1. No agents exist in the system → Lead is created unassigned; admin receives an alert to create agent accounts 2. Auto-assignment is disabled in settings → Lead remains with status 'New' in admin unassigned queue; admin manually assigns via UC-AD05 3. Assignment rule configuration is changed by Super Admin mid-operation → New rules apply to all subsequent leads; existing assignments are not changed
Post-conditions	• Lead is assigned to an agent (or to admin for manual handling) • Assignment timestamp is recorded • Escalation timer starts (UC-SY03): if agent does not contact within the configured window, escalation is triggered • Agent's lead count is incremented • Assignment event is visible in Admin Portal activity feed and Audit Trail

Use Case ID	UC-SY02
Use Case	Send Automated Notifications
Actor(s)	System (Automated — triggered by multiple use cases)
Description	The system sends automated email and SMS notifications to relevant actors when key platform events occur, ensuring timely communication without manual effort.
Pre-conditions	• Email (SMTP) and SMS (Twilio) credentials are configured in System Settings • Notification templates are configured for each event type • Recipient's contact details are available in the system

Main Flow	1. A triggering event occurs in the system (see Event Table below) 2. System identifies the notification type and recipient(s) for the event 3. System loads the corresponding email/SMS template 4. System populates template variables with event-specific data (e.g., customer name, property title, agent name, date) 5. System sends the email via the configured SMTP provider 6. System sends the SMS via Twilio API (if SMS is enabled for this event type) 7. System logs the notification: event type, recipient, channel, timestamp, delivery status 8. If delivery fails: System retries up to 3 times with exponential backoff 9. If all retries fail: System logs a notification failure alert in the Admin Portal
Alt / Exception Flows	1. Email delivery bounces (invalid email address) → Failure logged; admin sees 'Undeliverable Email' flag on the customer/user record 2. SMS fails (invalid phone number or insufficient credits) → SMS failure logged; email is still attempted as fallback 3. Recipient has unsubscribed from notifications → System checks opt-out flag before sending; notification is skipped
Post-conditions	• Notification is delivered to the recipient via the configured channel • Delivery status is logged (Sent / Delivered / Failed) • Failed notifications are flagged for admin review in the portal

Use Case ID	UC-SY03
Use Case	Escalate Uncontacted Lead
Actor(s)	System (Automated — scheduled job)
Description	The system automatically detects leads where the assigned agent has not logged any contact activity within the configured escalation window and alerts the Admin.
Pre-conditions	• A lead is assigned to an agent and has status 'Assigned' or 'New' • No call log or status update has been recorded against the lead • Escalation time window is configured in System Settings (default: 60 minutes from assignment)
Main Flow	1. System runs a scheduled background job every 15 minutes 2. Job queries all leads with status 'New' or 'Assigned' that have no activity logged since assignment 3. Job calculates time elapsed since assignment timestamp 4. If elapsed time exceeds the configured escalation window (e.g., 60 minutes): System flags the lead as 'Escalation Required' 5. System sends an escalation alert to Admin: Email + Portal notification with lead details and elapsed time 6. Lead appears in Admin Dashboard with an amber 'Escalation' badge 7. Admin investigates and takes action: contacts the agent, reassigns the lead (UC-AD05), or notes the delay reason 8. If agent logs a call or updates status before the window → escalation is not triggered for that lead
Alt / Exception Flows	1. Agent is on leave / unavailable → Admin reassigns the lead immediately to prevent customer experience delay 2. Escalation window is set to 0 (disabled) → System skips escalation checks for all leads 3. Multiple leads escalated simultaneously → Admin sees a list of all escalated leads filtered by age (oldest first) for priority handling
Post-conditions	• Admin is alerted to uncontacted leads and can take corrective action • No automatic reassignment occurs — escalation is informational only; human decision required

	● Escalation event is logged in the Activity Timeline of the affected lead ● Response time metric is captured for Agent Performance reporting
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6. CROSS-CUTTING CONCERNS

The following concerns apply across all use cases regardless of actor or module. They are handled at the infrastructure and middleware level and do not need to be repeated in each use case specification.

Concern	Implementation
Authentication	JWT-based session tokens for all portal users. Sessions expire after configurable idle timeout (default: 60 mins admin, 30 mins agent).
Authorization	All API endpoints enforce RBAC middleware server-side. Frontend role checks are supplementary only; backend is the authority.
Data Scoping	Agent sessions are scoped: all database queries for agent-role users automatically filter by Agent ID. No cross-agent data leakage.
Input Validation	All user-submitted forms validated on both client (instant feedback) and server (authoritative). Parameterized queries prevent SQL injection.
XSS Prevention	All user-generated content is sanitized before rendering. Rich text editor output is HTML-encoded. CSP headers enforced.
CSRF Protection	Anti-CSRF tokens required for all state-changing form submissions in admin and agent portals.
Rate Limiting	Login endpoints: max 5 attempts / 15 min per IP. Inquiry form: max 3 submissions / 10 min per IP to prevent lead spam.
Audit Trail	Every state-changing action (Create/Update/Delete) writes an immutable audit log entry. Logs cannot be edited or deleted.
Data Retention	Lead records retained indefinitely (configurable). Customer personal data can be exported or deleted on GDPR request.
Error Handling	All exceptions are caught; user-friendly error messages shown. Raw error details never exposed to client. Errors logged server-side.

7. USE CASE RELATIONSHIP SUMMARY

Include & Extend Relationships

«include» = the base use case always invokes the included use case as part of its flow. «extend» = the extending use case adds behaviour to the base use case only under certain conditions.

Base Use Case	Relationship	Related Use Case
UC-C06 Submit Inquiry	«include»	UC-SY01 Auto Lead Assignment
UC-C06 Submit Inquiry	«include»	UC-SY02 Send Notifications
UC-A03 Update Lead Status	«include»	UC-SY02 Send Notifications (on Won/key events)
UC-A05 Schedule Viewing	«include»	UC-SY02 Send Notifications (viewing confirmation)
UC-AD05 Assign Lead	«include»	UC-SY02 Send Notifications (agent alert)
UC-SY01 Auto Lead Assignment	«include»	UC-SY02 Send Notifications
UC-SY01 Auto Lead Assignment	«extend»	UC-SY03 Escalate Uncontacted Lead (if no contact)
UC-AD02 Manage Listings	«extend»	UC-AD03 Publish Listing (when ready)
UC-C01 Register Account	«extend»	UC-C06 Submit Inquiry (enables tracking)
UC-SA01 System Configuration	«include»	UC-SY01 Configures assignment rules
UC-AD07 Manage Roles & Permissions	«include»	All use cases (governs access)

Dependency Relationships

Use cases that depend on other use cases having run successfully first.

Use Case	Type	Depends On	Reason
UC-C06 Submit Inquiry	Depends on	UC-AD02 / UC-AD03	A published listing must exist for an inquiry to reference
UC-A02 View Assigned Leads	Depends on	UC-SY01	Leads appear only after auto-assignment runs
UC-AD09 View Reports	Depends on	UC-C06, UC-A03	Report data is populated by lead and agent activity
UC-AD10 Audit Logs	Depends on	All use cases	Every action in the system writes to the audit trail
UC-SY02 Notifications	Depends on	UC-SA01	Requires SMTP/SMS credentials configured in system settings
UC-SA02 Integrations	Depends on	UC-AD03	Portal sync is triggered by property publish event

Actor Interaction Summary

Actor	Initiates	Receives	Triggers
Customer	UC-C01 through UC-C08	Email confirmations, status updates	UC-SY01, UC-SY02

Actor	Initiates	Receives	Triggers
Agent	UC-A01 through UC-A08	Lead alerts, viewing reminders	UC-SY02 (on status change)
Admin	UC-AD01 through UC-AD10	Escalation alerts, report emails	UC-SY01 (assignment rules), UC-SY02
Super Admin	UC-SA01, UC-SA02, UC-AD01–AD10	System failure alerts	All system rules via configuration
System	UC-SY01, UC-SY02, UC-SY03	Event triggers from all other actors	Downstream notifications & escalations